

# LESSON 5: INTRODUCTION TO BERNOULLI'S THEOREM

Bernoulli's equation states that:

“In an ideal incompressible fluid when the flow is steady and continuous, the sum of pressure energy, kinetic energy and potential (or datum) energy is constant along a stream line.”

Mathematically,

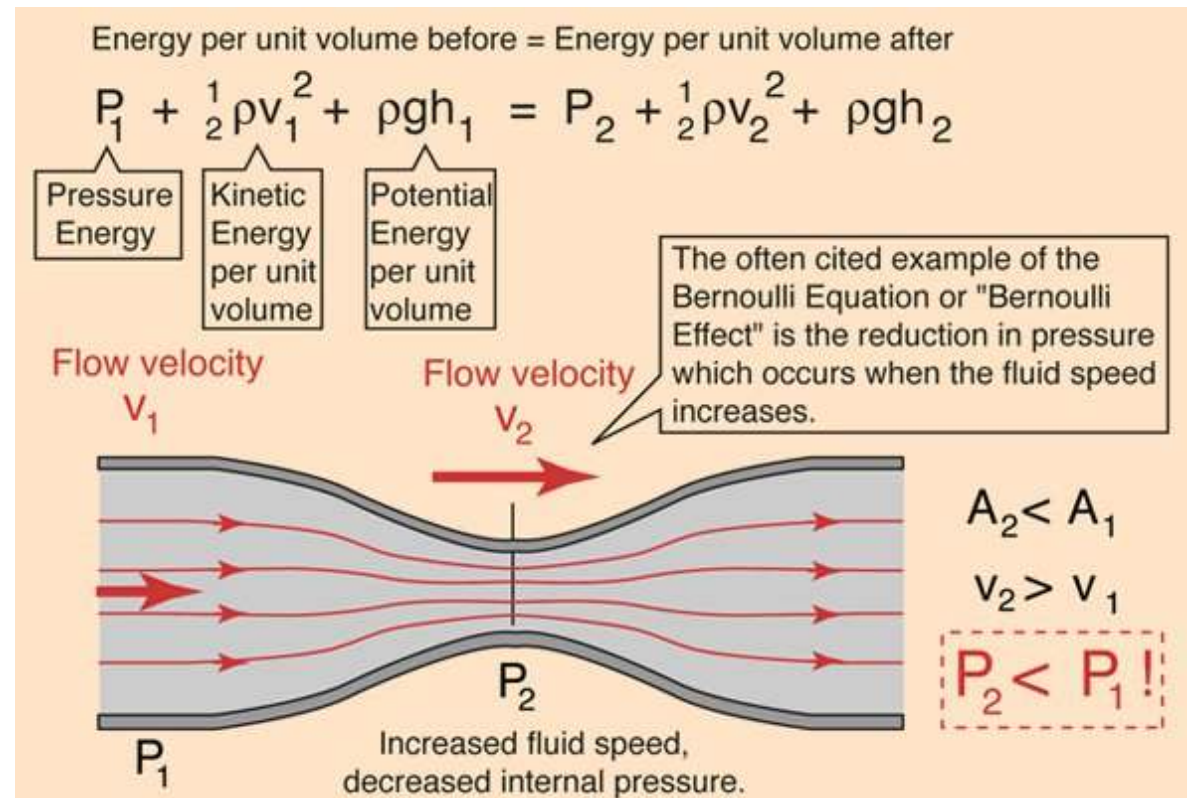
$$\frac{p}{w} + \frac{V^2}{2g} + z = \text{constant}$$

where,

$\frac{p}{w}$  = Pressure energy,

$\frac{V^2}{2g}$  = Kinetic energy, and

$z$  = Datum (or elevation) energy.



Bernoulli's equation can be obtained by integrating Euler's equation, where Euler's eq. is given as:

$$\frac{dp}{\rho} + g dz + v dv = 0$$

Thus;

$$\int \frac{dp}{\rho} + \int g dz + \int v dv = \text{constant}$$

If flow is incompressible,  $\rho$  is constant and

$$\therefore \frac{p}{\rho} + gz + \frac{v^2}{2} = \text{constant}$$

$$\text{or} \quad \frac{p}{\rho g} + z + \frac{v^2}{2g} = \text{constant}$$

$$\text{or} \quad \frac{p}{\rho g} + \frac{v^2}{2g} + z = \text{constant}$$

$\frac{p}{\rho g}$  = pressure energy per unit weight of fluid or pressure head.

$v^2/2g$  = kinetic energy per unit weight or kinetic head.

$z$  = potential energy per unit weight or potential head.

## Assumptions in deriving Bernouli's equation:

1. The fluid is ideal i.e viscosity is zero
2. The flow is steady
3. The flow is incompressible
4. The flow is irrotational

## Worked Example 1:

Water is flowing through a pipe of 5 cm diameter under pressure of 29.43 N/cm<sup>2</sup> (gauge) and with mean velocity of 2.0 m/s. Find the total head or total energy per unit weight of the water at a cross-section, which is 5 m above the datum line.

**Solution.** Given :

Diameter of pipe = 5 cm = 0.5 m

Pressure,  $p = 29.43 \text{ N/cm}^2 = 29.43 \times 10^4 \text{ N/m}^2$

Velocity,  $v = 2.0 \text{ m/s}$

Datum head,  $z = 5 \text{ m}$

Total head = pressure head + kinetic head + datum head

Pressure head  $= \frac{p}{\rho g} = \frac{29.43 \times 10^4}{1000 \times 9.81} = 30 \text{ m}$   $\left\{ \rho \text{ for water} = 1000 \frac{\text{kg}}{\text{m}^3} \right\}$

Kinetic head  $= \frac{v^2}{2g} = \frac{2 \times 2}{2 \times 9.81} = 0.204 \text{ m}$

$\therefore$  Total head  $= \frac{p}{\rho g} + \frac{v^2}{2g} + z = 30 + 0.204 + 5 = \mathbf{35.204 \text{ m. Ans.}}$

## Worked Example 2:

Water is flowing through a pipe of 20 cm diameter at one section and 10 cm at another section respectively. The rate of flow through the pipe is 35 litres/s. The section 1 is 6 m above the datum and section 2 is 4 m above the datum. If the pressure at section 1 is 39.24 N/cm<sup>2</sup>, find the intensity of pressure at section 2.

At section 1,

$$D_1 = 20 \text{ cm} = 0.2 \text{ m}$$

$$A_1 = \frac{\pi}{4} (.2)^2 = .0314 \text{ m}^2$$

$$p_1 = 39.24 \text{ N/cm}^2 \\ = 39.24 \times 10^4 \text{ N/m}^2$$

$$z_1 = 6.0 \text{ m}$$

At section 2,

$$D_2 = 0.10 \text{ m}$$

$$A_2 = \frac{\pi}{4} (0.1)^2 = .00785 \text{ m}^2$$

$$z_2 = 4 \text{ m}$$

$$p_2 = ?$$

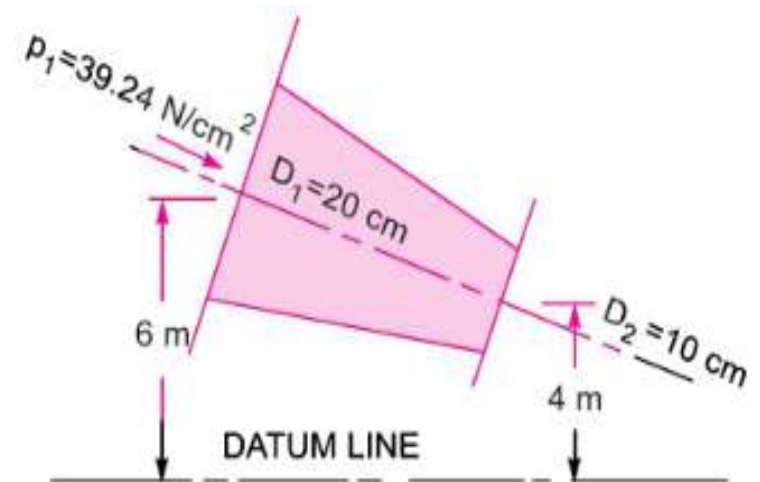
Rate of flow,

$$Q = 35 \text{ lit/s} = \frac{35}{1000} = .035 \text{ m}^3/\text{s}$$

Now

$$Q = A_1 V_1 = A_2 V_2$$

$$\therefore V_1 = \frac{Q}{A_1} = \frac{.035}{.0314} = 1.114 \text{ m/s}$$



and

$$V_2 = \frac{Q}{A_2} = \frac{.035}{.00785} = 4.456 \text{ m/s}$$

Applying Bernoulli's equation at sections 1 and 2, we get

$$\frac{p_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{V_2^2}{2g} + z_2$$

$$\text{or } \frac{39.24 \times 10^4}{1000 \times 9.81} + \frac{(1.114)^2}{2 \times 9.81} + 6.0 = \frac{p_2}{1000 \times 9.81} + \frac{(4.456)^2}{2 \times 9.81} + 4.0$$

$$\text{or } 40 + 0.063 + 6.0 = \frac{p_2}{9810} + 1.012 + 4.0$$

$$\text{or } 46.063 = \frac{p_2}{9810} + 5.012$$

$$\therefore \frac{p_2}{9810} = 46.063 - 5.012 = 41.051$$

$$\begin{aligned} \therefore p_2 &= 41.051 \times 9810 \text{ N/m}^2 \\ &= \frac{41.051 \times 9810}{10^4} \text{ N/cm}^2 = \mathbf{40.27 \text{ N/cm}^2}. \text{ Ans.} \end{aligned}$$

### Worked Example 3:

Water is flowing through a pipe of diameter 300 mm diameter and 200 mm at the bottom and upper end respectively. The intensity of pressure at the bottom end is 24.525 N/cm<sup>2</sup> and the pressure at the upper end is 9.81 N/cm<sup>2</sup>. Determine the difference in datum head if the rate of flow through the pipe is 40 litres/s

**Solution.** Given :

**Section 1,**  $D_1 = 300 \text{ mm} = 0.3 \text{ m}$   
 $p_1 = 24.525 \text{ N/cm}^2 = 24.525 \times 10^4 \text{ N/m}^2$

**Section 2,**  $D_2 = 200 \text{ mm} = 0.2 \text{ m}$   
 $p_2 = 9.81 \text{ N/cm}^2 = 9.81 \times 10^4 \text{ N/m}^2$

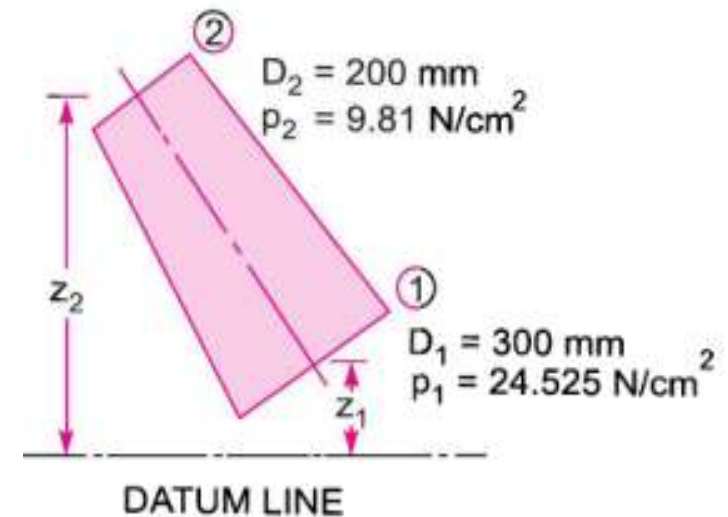
Rate of flow = 40 lit/s

$$Q = \frac{40}{1000} = 0.04 \text{ m}^3/\text{s}$$

$$A_1 V_1 = A_2 V_2 = \text{rate of flow} = 0.04$$

$$V_1 = \frac{.04}{A_1} = \frac{.04}{\frac{\pi}{4} D_1^2} = \frac{0.04}{\frac{\pi}{4} (0.3)^2} = 0.5658 \text{ m/s}$$

$$\simeq 0.566 \text{ m/s}$$



$$V_2 = \frac{.04}{A_2} = \frac{.04}{\frac{\pi}{4}(D_2)^2} = \frac{0.04}{\frac{\pi}{4}(0.2)^2} = 1.274 \text{ m/s}$$

Applying Bernoulli's equation at sections (1) and (2), we get

$$\frac{p_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{V_2^2}{2g} + z_2$$

$$\text{or } \frac{24.525 \times 10^4}{1000 \times 9.81} + \frac{.566 \times .566}{2 \times 9.81} + z_1 = \frac{9.81 \times 10^4}{1000 \times 9.81} + \frac{(1.274)^2}{2 \times 9.81} + z_2$$

$$\text{or } 25 + .32 + z_1 = 10 + 1.623 + z_2$$

$$\text{or } 25.32 + z_1 = 11.623 + z_2$$

$$\therefore z_2 - z_1 = 25.32 - 11.623 = 13.697 = 13.70 \text{ m}$$

$$\therefore \text{ Difference in datum head } = z_2 - z_1 = \mathbf{13.70 \text{ m. Ans.}}$$

## Worked Example 4:

Water is flowing through a tapered pipe of length 100 m having diameters 600 mm at the upper end and 300 mm at the lower end, at a rate of 50 litres/s. The pipe has a slope of 1 in 30. Find the pressure at the lower end if the pressure at the higher level is 19.62 N/cm<sup>2</sup>

**Solution.** Given :

Length of pipe,

$$L = 100 \text{ m}$$

Dia. at the upper end,

$$D_1 = 600 \text{ mm} = 0.6 \text{ m}$$

∴ Area,

$$\begin{aligned} A_1 &= \frac{\pi}{4} D_1^2 = \frac{\pi}{4} \times (.6)^2 \\ &= 0.2827 \text{ m}^2 \end{aligned}$$

$$\begin{aligned} p_1 &= \text{pressure at upper end} \\ &= 19.62 \text{ N/cm}^2 \end{aligned}$$

$$= 19.62 \times 10^4 \text{ N/m}^2$$

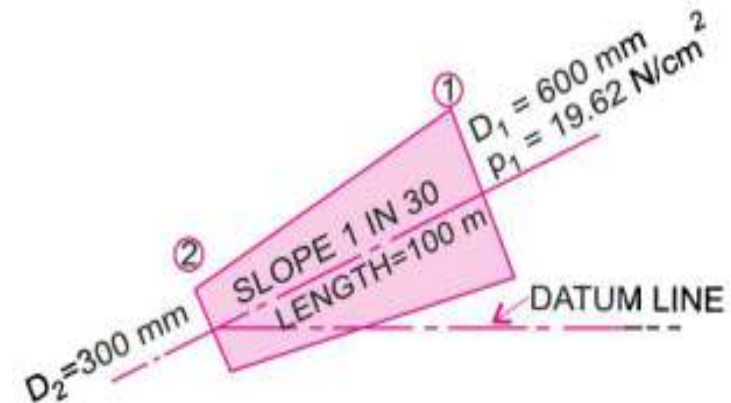
Dia. at lower end,

$$D_2 = 300 \text{ mm} = 0.3 \text{ m}$$

∴ Area,

$$A_2 = \frac{\pi}{4} D_2^2 = \frac{\pi}{4} (.3)^2 = 0.07068 \text{ m}^2$$

$$Q = \text{rate of flow} = 50 \text{ litres/s} = \frac{50}{1000} = 0.05 \text{ m}^3/\text{s}$$



Let the datum line passes through the centre of the lower end.

Then  $z_2 = 0$

As slope is 1 in 30 means  $z_1 = \frac{1}{30} \times 100 = \frac{10}{3} \text{ m}$

Also we know  $Q = A_1 V_1 = A_2 V_2$

$\therefore V_1 = \frac{Q}{A} = \frac{0.05}{.2827} = 0.1768 \text{ m/sec} = 0.177 \text{ m/s}$

and  $V_2 = \frac{Q}{A_2} = \frac{0.5}{.07068} = 0.7074 \text{ m/sec} = 0.707 \text{ m/s}$

Applying Bernoulli's equation at sections (1) and (2), we get

$$\frac{p_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{V_2^2}{2g} + z_2$$

$$\text{or } \frac{19.62 \times 10^4}{1000 \times 9.81} + \frac{.177^2}{2 \times 9.81} + \frac{10}{3} = \frac{p_2}{\rho g} + \frac{.707^2}{2 \times 9.81} + 0$$

$$\text{or } 20 + 0.001596 + 3.334 = \frac{p_2}{\rho g} + 0.0254$$

$$\text{or } 23.335 - 0.0254 = \frac{p_2}{1000 \times 9.81}$$

$$\text{or } p_2 = 23.3 \times 9810 \text{ N/m}^2 = 228573 \text{ N/m}^2 = \mathbf{22.857 \text{ N/cm}^2}. \text{ Ans.}$$

## Bernoulli's Equation for Real Fluid

Bernoulli's equation was derived on the assumption that fluid is inviscid (non-viscous) and therefore frictionless. But all the real fluids are viscous and hence offer resistance to flow. Thus, there are always some losses in fluid flows and hence in the application of Bernoulli's equation, these losses have to be taken into consideration. Thus, the Bernoulli's equation between points 1 and 2 is given as:

$$\frac{p_1}{\rho g} + \frac{v_1^2}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{v_2^2}{2g} + z_2 + h_L$$

where  $h_L$  is loss of energy between points 1 and 2.

## Worked Example 5:

A pipe of diameter 400 mm carries water at a velocity of 25 m/s. The pressures at the points A and B are given as 29.43 N/cm<sup>2</sup> and 22.563 N/cm<sup>2</sup> respectively while the datum head at A and B are 28 and 30 m. Find the loss of head between A and B.

**Solution.** Given :

Dia. of pipe,  $D = 400 \text{ mm} = 0.4 \text{ m}$

Velocity,  $V = 25 \text{ m/s}$

At point A,  $p_A = 29.43 \text{ N/cm}^2 = 29.43 \times 10^4 \text{ N/m}^2$

$z_A = 28 \text{ m}$

$v_A = v = 25 \text{ m/s}$

$\therefore$  Total energy at A,  $E_A = \frac{p_A}{\rho g} + \frac{v_A^2}{2g} + z_A$

$$= \frac{29.43 \times 10^4}{1000 \times 9.81} + \frac{25^2}{2 \times 9.81} + 28$$

$$= 30 + 31.85 + 28 = 89.85 \text{ m}$$

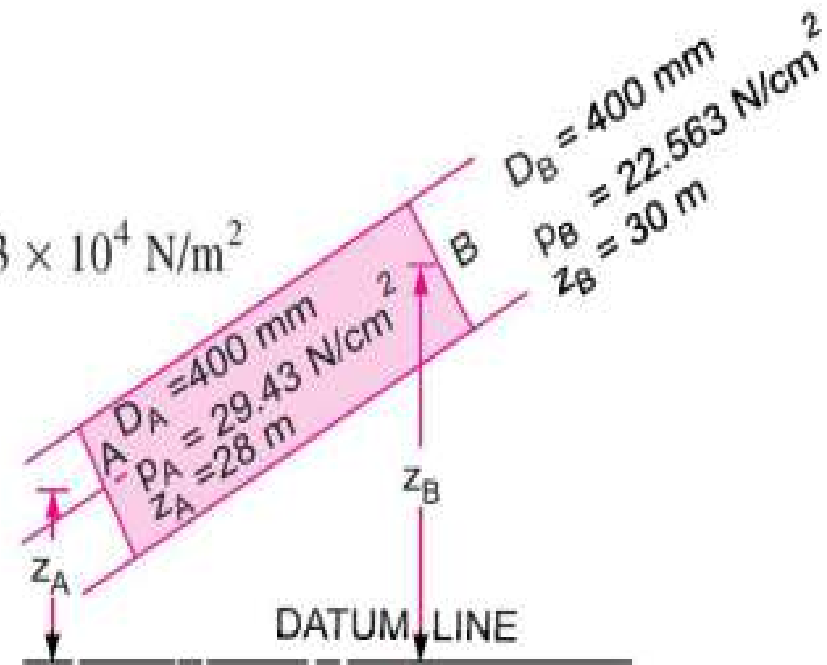


Fig. 6.6

**At point  $B$ ,**

$$p_B = 22.563 \text{ N/cm}^2 = 22.563 \times 10^4 \text{ N/m}^2$$

$$z_B = 30 \text{ m}$$

$$v_B = v = v_A = 25 \text{ m/s}$$

**$\therefore$  Total energy at  $B$ ,**

$$E_B = \frac{p_B}{\rho g} + \frac{v_B^2}{2g} + z_B$$

$$= \frac{22.563 \times 10^4}{1000 \times 9.81} + \frac{25^2}{2 \times 9.81} + 30 = 23 + 31.85 + 30 = 84.85 \text{ m}$$

**$\therefore$  Loss of energy**

$$= E_A - E_B = 89.85 - 84.85 = \mathbf{5.0 \text{ m. Ans.}}$$

## Worked Example 6:

A pipeline carrying oil of specific gravity 0.87, changes in diameter from 200 mm at position A to 500 mm at position B which is 4 metres at a higher level. If the pressures at A and B are 9.81 N/cm<sup>2</sup> and 5.886 N/cm<sup>2</sup> respectively and the discharge is 200 litres/s. determine the loss of head and direction of flow.

**Solution.** Discharge,  $Q = 200 \text{ lit/s} = 0.2 \text{ m}^3/\text{s}$

Sp. gr. of oil  $= 0.87$

$\therefore \rho \text{ for oil} = 0.87 \times 1000 = 870 \frac{\text{kg}}{\text{m}^3}$

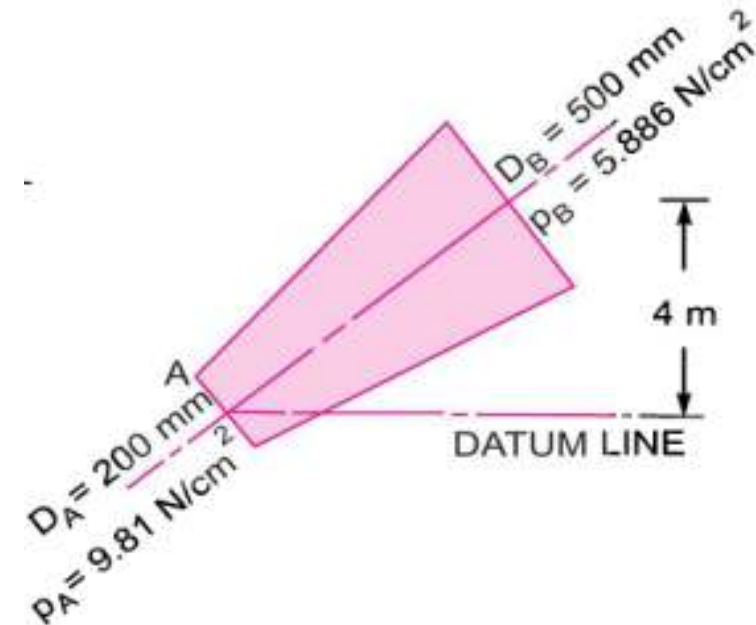
Given : At section A,  $D_A = 200 \text{ mm} = 0.2 \text{ m}$

Area,  $A_A = \frac{\pi}{4} (D_A)^2 = \frac{\pi}{4} (.2)^2$   
 $= 0.0314 \text{ m}^2$   
 $p_A = 9.81 \text{ N/cm}^2$   
 $= 9.81 \times 10^4 \text{ N/m}^2$

If datum line is passing through A, then

$$Z_A = 0$$

$$V_A = \frac{Q}{A_A} = \frac{0.2}{0.0314} = 6.369 \text{ m/s}$$



At section  $B$ ,

$$D_B = 500 \text{ mm} = 0.50 \text{ m}$$

Area,

$$A_B = \frac{\pi}{4} D_B^2 = \frac{\pi}{4} (.5)^2 = 0.1963 \text{ m}^2$$

$$p_B = 5.886 \text{ N/cm}^2 = 5.886 \times 10^4 \text{ N/m}^2$$

$$Z_B = 4.0 \text{ m}$$

$$V_B = \frac{Q}{\text{Area}} = \frac{0.2}{.1963} = 1.018 \text{ m/s}$$

Total energy at  $A$

$$= E_A = \frac{p_A}{\rho g} + \frac{V_A^2}{2g} + Z_A$$

$$= \frac{9.81 \times 10^4}{870 \times 9.81} + \frac{(6.369)^2}{2 \times 9.81} + 0 = 11.49 + 2.067 = 13.557 \text{ m}$$

Total energy at  $B$

$$= E_B = \frac{p_B}{\rho g} + \frac{V_B^2}{2g} + Z_B$$

$$= \frac{5.886 \times 10^4}{870 \times 9.81} + \frac{(1.018)^2}{2 \times 9.81} + 4.0 = 6.896 + 0.052 + 4.0 = 10.948 \text{ m}$$

(i) **Direction of flow.** As  $E_A$  is more than  $E_B$  and hence flow is taking place from  $A$  to  $B$ . **Ans.**

(ii) Loss of head  $= h_L = E_A - E_B = 13.557 - 10.948 = \mathbf{2.609 \text{ m}}$ . **Ans.**

# Application of Bernoulli's Equation

Bernoulli's equation applies to all problems of incompressible fluid flow where energy considerations are involved. However, in this scope, we consider its application in the following devices:

1. Venturi meter
2. Orifice meter
3. Pilot – tube

## Venturimeter

A venturimeter is a device used for measuring the rate of flow of a fluid flowing through a pipe. It consists of the following parts:

- i). A short converging part      ii) Throat and      iii) Diverging part

Fig 6.9 illustrates a venturimeter say through which water flows.

Let  $d_1$  = diameter at inlet or at section (1),

$p_1$  = pressure at section (1)

$v_1$  = velocity of fluid at section (1),

$$a = \text{area at section (1)} = \frac{\pi}{4} d_1^2$$

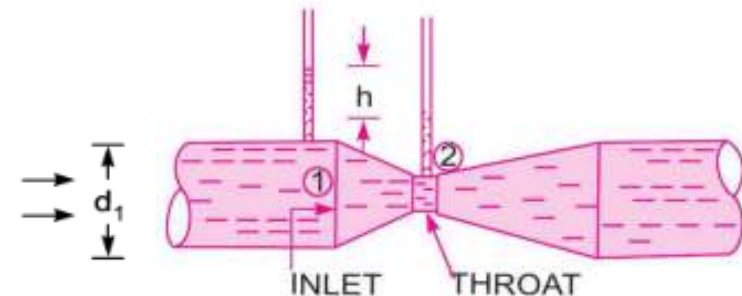


Fig. 6.9 Venturimeter.

and  $d_2, p_2, v_2, a_2$  are corresponding values at section (2).

Applying Bernoulli's equation at sections (1) and (2), we get

$$\frac{p_1}{\rho g} + \frac{v_1^2}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{v_2^2}{2g} + z_2$$

As pipe is horizontal, hence  $z_1 = z_2$

$$\frac{p_1}{\rho g} + \frac{v_1^2}{2g} = \frac{p_2}{\rho g} + \frac{v_2^2}{2g} \quad \text{or} \quad \frac{p_1 - p_2}{\rho g} = \frac{v_2^2}{2g} - \frac{v_1^2}{2g}$$

But  $\frac{p_1 - p_2}{\rho g}$  is the difference of pressure heads at sections 1 and 2 and it is equal to  $h$  or  $\frac{p_1 - p_2}{\rho g} = h$

Substituting this value of  $\frac{p_1 - p_2}{\rho g}$  in the above equation, we get

$$h = \frac{v_2^2}{2g} - \frac{v_1^2}{2g}$$

Now applying continuity equation at sections 1 and 2

$$a_1 v_1 = a_2 v_2 \quad \text{or} \quad v_1 = \frac{a_2 v_2}{a_1}$$

Substituting this value of  $v_1$  in equation

$$h = \frac{v_2^2}{2g} - \frac{\left(\frac{a_2 v_2}{a_1}\right)^2}{2g} = \frac{v_2^2}{2g} \left[1 - \frac{a_2^2}{a_1^2}\right] = \frac{v_2^2}{2g} \left[\frac{a_1^2 - a_2^2}{a_1^2}\right]$$

or

$$v_2^2 = 2gh \frac{a_1^2}{a_1^2 - a_2^2}$$

$\therefore$

$$v_2 = \sqrt{2gh \frac{a_1^2}{a_1^2 - a_2^2}} = \frac{a_1}{\sqrt{a_1^2 - a_2^2}} \sqrt{2gh}$$

$\therefore$  Discharge,

$$\begin{aligned} Q &= a_2 v_2 \\ &= a_2 \frac{a_1}{\sqrt{a_1^2 - a_2^2}} \times \sqrt{2gh} = \frac{a_1 a_2}{\sqrt{a_1^2 - a_2^2}} \times \sqrt{2gh} \end{aligned}$$

The equation above gives the discharge under ideal conditions and is called , theoretical discharge. Actual discharge will be less than theoretical discharge and is expressed as:

$$Q_{\text{act}} = C_d \times \frac{a_1 a_2}{\sqrt{a_1^2 - a_2^2}} \times \sqrt{2gh}$$

where  $C_d$  = Co-efficient of venturimeter and its value is less than 1.

The value of “h” can be given by differential U-tube manometer and if

**I.** The differential manometer contain a liquid heavier than the liquid flowing through the pipe, Let:

$S_h$  = Sp. gravity of the heavier liquid

$S_o$  = Sp. gravity of the liquid flowing through pipe

$x$  = Difference of the heavier liquid column in U-tube

$$h = x \left[ \frac{S_h}{S_o} - 1 \right]$$

**II.** The differential manometer contain a liquid lighter than the liquid flowing through the pipe, h is given by:

$$h = x \left[ 1 - \frac{S_l}{S_o} \right]$$

where  $S_l$  = Sp. gr. of lighter liquid in U-tube

$S_o$  = Sp. gr. of fluid flowing through pipe

$x$  = Difference of the lighter liquid columns in U-tube.

**III.** Inclined venturimeter with differential U-tube manometer. This case is related to inclined venturimeter having differential U-tube manometer. So let the differential manometer contain heavier liquid, then h is given as;

$$h = \left( \frac{p_1}{\rho g} + z_1 \right) - \left( \frac{p_2}{\rho g} + z_2 \right) = x \left[ \frac{S_h}{S_o} - 1 \right]$$

**IV.** Now consider an inclined venturimeter in which differential manometer contains a liquid which is lighter than the liquid flowing through the pipe. The value of h is given as:

$$h = \left( \frac{p_1}{\rho g} + z_1 \right) - \left( \frac{p_2}{\rho g} + z_2 \right) = x \left[ 1 - \frac{S_l}{S_o} \right]$$

## Worked Example 7:

A horizontal venturimeter with inlet and throat diameters 30 cm and 15 cm respectively is used to measure the flow of water. The reading of differential manometer connected to the inlet and the throat is 20 cm of mercury. Determine the rate of flow. Take  $C_d = 0.98$ .

**Solution.** Given :

Dia. at inlet,

$$d_1 = 30 \text{ cm}$$

∴ Area at inlet,

$$a_1 = \frac{\pi}{4} d_1^2 = \frac{\pi}{4} (30)^2 = 706.85 \text{ cm}^2$$

Dia. at throat,

$$d_2 = 15 \text{ cm}$$

∴

$$a_2 = \frac{\pi}{4} \times 15^2 = 176.7 \text{ cm}^2$$

$$C_d = 0.98$$

Reading of differential manometer =  $x = 20$  cm of mercury.

∴ Difference of pressure head is given by (6.9)

or

$$h = x \left[ \frac{S_h}{S_o} - 1 \right]$$

where  $S_h$  = Sp. gravity of mercury = 13.6,  $S_o$  = Sp. gravity of water = 1

$$= 20 \left[ \frac{13.6}{1} - 1 \right] = 20 \times 12.6 \text{ cm} = 252.0 \text{ cm of water.}$$

The discharge through venturimeter is given by eqn. (6.8)

$$\begin{aligned} Q &= C_d \frac{a_1 a_2}{\sqrt{a_1^2 - a_2^2}} \times \sqrt{2gh} \\ &= 0.98 \times \frac{706.85 \times 176.7}{\sqrt{(706.85)^2 - (176.7)^2}} \times \sqrt{2 \times 9.81 \times 252} \end{aligned}$$

$$\begin{aligned} &= \frac{86067593.36}{\sqrt{499636.9 - 31222.9}} = \frac{86067593.36}{684.4} \\ &= 125756 \text{ cm}^3/\text{s} = \frac{125756}{1000} \text{ lit/s} = \mathbf{125.756 \text{ lit/s. Ans.}} \end{aligned}$$

## Worked Example 8:

A horizontal venturimeter with inlet diameters 20 cm and throat diameter 10 cm is used to measure the flow of oil of sp gr 0.8. The discharge of oil through the venturimeter is 60 litres/s. Find the reading of the oil-mercury differential manometer.. Take  $C_d = 0.98$ .

**Solution.** Given :

$$d_1 = 20 \text{ cm}$$
$$\therefore a_1 = \frac{\pi}{4} 20^2 = 314.16 \text{ cm}^2$$
$$d_2 = 10 \text{ cm}$$
$$\therefore a_2 = \frac{\pi}{4} \times 10^2 = 78.54 \text{ cm}^2$$
$$C_d = 0.98$$
$$Q = 60 \text{ litres/s} = 60 \times 1000 \text{ cm}^3/\text{s}$$

Using the equation (6.8),

$$Q = C_d \frac{a_1 a_2}{\sqrt{a_1^2 - a_2^2}} \times \sqrt{2gh}$$
$$60 \times 1000 = 0.98 \times \frac{314.16 \times 78.54}{\sqrt{(314.16)^2 - (78.54)^2}} \times \sqrt{2 \times 981 \times h}$$
$$= \frac{1071068.78 \sqrt{h}}{304}$$
$$\sqrt{h} = \frac{304 \times 60000}{1071068.78} = 17.029$$
$$h = (17.029)^2 = 289.98 \text{ cm of oil}$$

$$\text{But, } h = x \left[ \frac{S_h}{S_o} - 1 \right]$$

where  $S_h = \text{Sp. gr. of mercury} = 13.6$   
 $S_o = \text{Sp. gr. of oil} = 0.8$   
 $x = \text{Reading of manometer}$

Therefore

$$289.98 = x \left[ \frac{13.6}{0.8} - 1 \right] = 16x$$
$$x = \frac{289.98}{16} = 18.12 \text{ cm.}$$

Therefore

Reading for oil-mercury  
manometer = **18.12 cm. Ans**

## Worked Example 9:

A horizontal venturimeter with inlet diameters 20 cm and throat diameter 10 cm is used to measure the flow of water. The pressure at inlet is 17.658 N/cm<sup>2</sup> and the vacuum pressure at the throat is 30 cm of mercury. Find the discharge of water through venturimeter. Take  $C_d = 0.98$ .

**Solution.** Given :

Dia. at inlet,

$$d_1 = 20 \text{ cm}$$

∴

$$a_1 = \frac{\pi}{4} \times (20)^2 = 314.16 \text{ cm}^2$$

Dia. at throat,

$$d_2 = 10 \text{ cm}$$

∴

$$a_2 = \frac{\pi}{4} \times 10^2 = 78.74 \text{ cm}^2$$

$$p_1 = 17.658 \text{ N/cm}^2 = 17.658 \times 10^4 \text{ N/m}^2$$

$\rho$  for water

$$= 1000 \frac{\text{kg}}{\text{m}^3} \text{ and } \therefore \frac{p_1}{\rho g} = \frac{17.658 \times 10^4}{9.81 \times 1000} = 18 \text{ m of water}$$

$$\frac{p_2}{\rho g} = -30 \text{ cm of mercury}$$

$$= -0.30 \text{ m of mercury} = -0.30 \times 13.6 = -4.08 \text{ m of water}$$

∴ Differential head

$$= h = \frac{p_1}{\rho g} - \frac{p_2}{\rho g} = 18 - (-4.08)$$

$$= 18 + 4.08 = 22.08 \text{ m of water} = 2208 \text{ cm of water}$$

The discharge  $Q$  is given as:

$$Q = C_d \frac{a_1 a_2}{\sqrt{a_1^2 - a_2^2}} \times \sqrt{2gh}$$

$$= 0.98 \times \frac{314.16 \times 78.54}{\sqrt{(314.16)^2 - (78.74)^2}} \times \sqrt{2 \times 9.81 \times 2208}$$

$$= \frac{50328837.21}{304} \times 165555 \text{ cm}^3/\text{s}$$

$$= \mathbf{165.555 \text{ lit/s. Ans.}}$$

## Worked Example 10:

A 20 cm x 10 cm venturimeter is inserted in a vertical pipe carrying oil of sp gr. 0.8, flowing in the upward direction. The difference of levels between the throat and inlet section is 50 cm. The oil mercury differential manometer gives a reading of 30 cm of mercury. Find the discharge of oil. Neglect losses.

**Solution.** Dia. at inlet,  $d_1 = 20$  cm

$$\therefore a_1 = \frac{\pi}{4} (20)^2 = 314.16 \text{ cm}^2$$

Dia. at throat,  $d_2 = 10$  cm

$$\therefore a_2 = \frac{\pi}{4} (10)^2 = 78.54 \text{ cm}^2$$

Sp. gr. of oil,  $S_o = 0.8$

Sp. gr. of mercury,  $S_g = 13.6$

Differential manometer reading,  $x = 30$  cm

$$\begin{aligned}\therefore h &= \left( \frac{p_1}{\rho g} + z_1 \right) - \left( \frac{p_2}{\rho g} + z_2 \right) = x \left[ \frac{S_g}{S_o} - 1 \right] \\ &= 30 \left[ \frac{13.6}{0.8} - 1 \right] = 30 [17 - 1] = 30 \times 16 = 480 \text{ cm of oil} \\ C_d &= 1.0\end{aligned}$$

The discharge,

$$\begin{aligned} Q &= C_d \frac{a_1 a_2}{\sqrt{a_1^2 - a_2^2}} \times \sqrt{2gh} \\ &= \frac{1.0 \times 314.16 \times 78.54}{\sqrt{(314.16)^2 - (78.54)^2}} \times \sqrt{2 \times 981 \times 480} \text{ cm}^3/\text{s} \\ &= \frac{23932630.7}{304} = 78725.75 \text{ cm}^3/\text{s} = \mathbf{78.725 \text{ litres/s. Ans.}} \end{aligned}$$

**THE END**